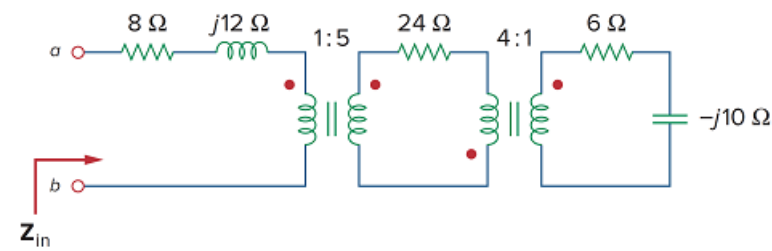


## Problem

Calculate the input impedance for the network in Fig. 13.115.

Figure 13.115



## Step-by-step solution

## Step 1 of 1

The value of  $Z_{in}$  is not affected by the location of the dots since  $n^2$  is involved.

$$Z'_{in} = \frac{(6 - j10)}{(n')^2}, \quad n' = \frac{1}{4}$$

$$Z'_{in} = 16(6 - j10) = 96 - j160$$

$$Z_{in} = 8 + j12 + \frac{(Z'_{in} + 24)}{n^2}, \quad n = 5$$

$$Z_{in} = 8 + j12 + \frac{(120 - j160)}{25}$$

$$= 8 + j12 + 4.8 - j6.4$$

$$Z_{in} = (12.8 + j5.6) \text{ Ohms}.$$